

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	
Project Title	A comprehensive measure of well being
Team Members	Chikali Sarika, Anusha Gollapalli, Karishma Sharon Kaparapu, Ambika Chitikela, Parneeth Devanaboina
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

This proposal outlines a machine learning solution that classifies countries into Human Development Index tiers based on life expectancy, education, and income indicators. It replaces slow, manual HDI interpretation with an interactive tool that gives policymakers, researchers, and development organizations instant, data-driven insight into a country's development level and the factors driving it.

Project Overview	
Objective	To build a machine learning model that classifies countries into Human Development Index tiers (Very High, High, Medium, Low) using life expectancy, education, and income indicators, giving policymakers and researchers a fast, interactive way to assess development levels.
Scope	The project covers data collection and preprocessing of HDI-related indicators, training and evaluating a classification model, and deploying it through a Flask web application where users can input indicator values for a country and instantly receive its predicted HDI tier.
Problem Statement	
Description	Computing and interpreting the Human Development Index currently relies on static annual reports and manual cross-referencing of health, education, and income data, making it slow for policymakers and researchers to test "what-if" scenarios or compare emerging economies in real time.
Impact	Solving this gives governments, researchers, and development organizations an instant, data-driven way to identify development gaps, prioritize interventions in healthcare, education, or income generation, and support faster, more informed policy decisions.
Proposed Solution	
Approach	Train a supervised machine learning classification model on historical country-level HDI data (life expectancy, mean years of schooling, expected years of schooling, and GNI per capita) to predict a country's HDI tier, and expose it through an interactive Flask web interface.
Key Features	- Machine learning-based classification model that predicts a country's HDI tier (Very High/High/Medium/Low). - Interactive Flask web app for entering indicator values and receiving instant predictions. - Support for multiple scenarios: high-performing nations, mid-range developing economies, and countries needing development intervention. - Data visualizations (Matplotlib, Seaborn) highlighting which indicators drive the predicted classification. - Reusable pipeline for retraining the model as new country indicator data becomes available.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications, number of cores	Standard CPU (Intel i5/i7 or equivalent); GPU not required for classical ML models
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	5 GB SSD
Software		
Frameworks	Python web framework	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Anaconda, Jupyter Notebook, PyCharm
Data		
Data	Source, size, format	UNDP Human Development Report indicators / Kaggle HDI dataset, country-level records, csv